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Incremental Object Detection via Future-Aware Decoupled Cross-Head Distillation

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Abstract

Incremental Object Detection (IOD) enables AI systems to continuously acquire new object classes while preserving knowledge of previously learned ones, an ability essential for deployment in dynamic, real-world environments. Existing IOD methods typically rely on knowledge distillation to mitigate catastrophic forgetting. However, the tight coupling between the student model's detection head and backbone causes distillation gradients to conflict with new-class supervision at the head, injecting head-specific bias into the backbone and ultimately weakening distillation effectiveness. To address this issue, we propose a decoupled training mechanism for the model's backbone and classification head. Specifically, we introduce the Future-aware Cross-head decoupled Distillation (FaCHD) method, which utilizes two frozen complementary teachers (historical and intermediate teachers) to decode the student's ROI features for cross-head distillation. This strategy implicitly alleviates prediction conflicts caused by detection-head bias and provides richer task-relevant guidance, thereby improving distillation efficiency. To further address the detection head bias and model recency problem, we propose a Prototype Semantic Drift Compensation module, which recalibrates multi-granularity prototypes of old classes, effectively correcting semantic drift and enhancing the stability of the detection head. Extensive experiments on two standard IOD benchmarks demonstrate the effectiveness and superiority of the proposed method.

1. Introduction

Object detection is a fundamental task in computer vision with critical applications in autonomous driving, robotics, and video understanding [9, 10, 35]. Conventional object detectors assume access to all categories and exhaustively annotated data at once, enabling static joint training. However, real-world deployment often involves dynamic environments where new object classes continually emerge, and storing historical data is impractical or restricted due to memory constraints and privacy concerns. These challenges have motivated the study of Incremental Object Detection (IOD) [31], which aims to continuously learn new classes while retaining previously acquired knowledge, effectively mitigating catastrophic forgetting in evolving, real-world scenarios.

Following the traditional Class Incremental Learning (CIL) paradigm, existing Incremental Object Detection (IOD) methods can be roughly categorized into regularization-based and replay-based approaches. Among regularization-based methods, Knowledge Distillation (KD) [5, 8, 11, 13, 23, 29, 31, 46] plays a vital role in mitigating catastrophic forgetting for IOD. Compared with CIL, IOD faces unique challenges arising from the co-occurrence of multiple task classes within the current training data: (1) foreground objects from previous tasks may remain unlabeled in the current stage, making them prone to misclassification as background; (2) the current-stage background may contain objects from future tasks, which may inadvertently be treated as foreground. These factors induce inter-class overlap and foreground-background confusion, further exacerbating cross-task interference during knowledge distillation. Although existing KD-based IOD methods impose strong regularization to prevent forgetting, they often overlook prediction conflicts arising from distillation in object detection models. Specifically, when distillation is applied directly to output logits, conflicts arise between the student model's assigner-based ground-truth targets and the teacher model's predictions. Moreover, many methods tightly couple the detection head and backbone, causing head-specific bias (typically biased toward new classes) to be imprinted into the backbone features, which further aggravates catastrophic forgetting.

To address these challenges, we propose Future-aware Decoupled Cross-Head Distillation (FaCHD), a novel framework for IOD that performs feature distillation

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through decoupled cross-head knowledge transfer from both past and future detection models. Unlike conventional approaches that directly enforce consistency between intermediate feature maps or teacher-student predictions, FaCHD explicitly conducts feature distillation under the regularization of past and future detection heads, thereby implicitly mitigating prediction conflicts arisen by the detection-head bias and improving distillation efficiency. Specifically, the intermediate features from the student's backbone are fed to both the past and future detection head to produce cross-head predictions [36]. Then, prediction concatenation and Knowledge distillation are applied between these cross-head predictions and the teachers' outputs. The key innovation of FaCHD lies in its cross-head decoding strategy: the student's Region of Interest (ROI) features are decoded by two teachers, each specialized in their respective knowledge domains, $i.e.$, previous and current category spaces. By decoupling the student's detection head from the backbone, gradients are constrained to flow exclusively through the backbone and ROI, ensuring stable, head-independent geometric representations, so as to address the detection haed bias and model recency problem. This strategy not only enhances training stability for prediction mimicking, but also provides more task-oriented information than conventional feature imitation.

To further address the detection head bias and model recency problem, existing class incremental learning method usually store prototypes/statistics for old categories, faciliating the joint learning with current class features. However, features of previously learned classes may become misaligned due to semantic drift, which can reduce the efficiency of detection head learning. To this end, we propose a semantic drift compensation mechanism over multi-granularity ROI prototypes. Specifically, we maintain a bank of ROI prototypes at multiple granularities and estimate class-specific semantic offsets by comparing feature maps from the current (student) and historical models. Recalibrating these prototypes allows the model to automatically correct drift without requiring annotations for old classes. This mechanism operates primarily on the student's detection head, resetting task-specific decision boundaries while preserving the underlying geometric consistency of features. By separately addressing the tasks of backbone geometric consistency shaping through FaCHD and head decision boundary resetting through prototype compensation, our framework systematically and modularly mitigates catastrophic forgetting, enhancing both stability and adaptability in incremental object detection.

To summarize, our contributions are as follows:

- We propose a future-aware decoupled cross-Head distillation mechanism for IOD, which performs feature distillation through decoupled cross-head knowledge transfer from both past and current detection models. This strategy not only enhances training stability for prediction mimicking, but also provides more task-oriented information than conventional feature imitation.
- We propose Prototype Semantic Drift Compensation mechanism, which recalibrates multi-granularity prototypes for old classes, solving the problem of semantic drift and resetting task-specific decision boundary of the detection head.
- Through extensive experiments, we validate the effectiveness of our approach in reducing catastrophic forgetting across two benchmark IOD datasets, achieving superior performance compared to state-of-the-art methods.

2. Related Work

Class Incremental Learning. Existing class-incremental learning methods can be broadly grouped into three categories: (1) regularization- and distillation-based approaches, (2) replay- and prototype-based methods, and (3) parameter-isolation or dynamic-architecture strategies. Our work is most closely related to the knowledge distillation–based line, which remains the dominant paradigm in this field. LwF [19] preserves prior knowledge by aligning the student's predictions with a frozen teacher, while EWC [17] restricts updates to parameters identified as important through Fisher information. Subsequent studies further refine distillation using adaptive regularization, and task-aware feature or logit alignment [1, 2, 6, 18, 21, 26, 27, 44]. Despite these advances, distillation remains highly sensitive to conflicts between old and new tasks, a challenge our method aims to address.

Incremental Object Detection. Compared with class-incremental learning, incremental object detection (IOD) is more challenging because the model must preserve spatial localization while coping with missing annotations and background conflicts, $i.e.$, unlabeled old or future objects are often treated as background, causing severe supervision bias. Within two-stage detectors such as Faster R-CNN also used in our work, ILOD [34] first introduced knowledge distillation into IOD. Follow-up methods further explored where and how to distill within the detection pipeline. Faster ILOD [32], AFD [22], and ERD [11] strengthen the preservation of old knowledge across the RPN, ROI features, and the classification and regression heads. To mitigate non-co-occurrence between old and new classes, dual-teacher frameworks exploit unlabeled in-the-wild data and fused teacher probabilities to reduce information asymmetry [8, 30, 45]. Rehearsal-based methods have also evolved: Augmented Box Replay (ABR) [25] replays only foreground boxes using mixup or mosaic augmentation, avoiding the foreground/background shift caused by full-image replay. More recently, prototype- and distribution-alignment techniques have been explored. For example, Gaussian-mixture modeling of high-dimensional prototypes

captures both intra- and inter-class variability across stages, helping alleviate semantic drift in the detector head [38].

3. Methodology

3.1. Preliminaries

Problem Formulation. Incremental object detection trains a detector over N successive stages, each introducing a fresh subset of categories. Let the full category space be $\mathcal{C} = \{\mathcal{C}_1, \ldots, \mathcal{C}_t, \ldots, \mathcal{C}_N\}$, where each subset $\mathcal{C}_t$ represents a disjoint set of categories introduced at stage t, with the label space progressively growing as new categories are added at each stage. At stage t, the data $\mathcal{D}_t = \{\mathcal{X}_t, \mathcal{Y}_t\}$ include images and annotations. An image $\mathbf{x}_i \in \mathcal{X}_t$ may contain objects from multiple classes in $\mathcal{C}$, but only instances belonging to $\mathcal{C}_t$ are annotated. The goal is to update the detector from $\mathcal{M}_{t-1}$ to $\mathcal{M}_t$ using only $\mathcal{D}_t$, without access to prior datasets $\mathcal{D}_{1:t-1}$, while maintaining or improving performance on previously learned classes.

General Object Detector. Following prior work [4, 24, 29, 42], we adopt Faster R-CNN as the base detection framework, which is trained end-to-end and consists of three modules: a backbone feature extractor $\mathcal{M}_f$, a Region Proposal Network (RPN) $\mathcal{M}_{\text{rpn}}$, and a region-wise prediction head (ROI head) $\mathcal{M}_{\text{roi}}$, with the classification head denoted by $\mathcal{H}$. Given an input image, $\mathcal{M}_f$ produces a dense feature map. On top of these features, $\mathcal{M}_{\text{rpn}}$ evaluates predefined anchors at each spatial location and regresses bounding-box offsets $\mathbf{l} \in \mathbb{R}^{1\times 4}$ to form proposals. The head $\mathcal{M}_{\text{roi}}$ then takes both positive and negative proposals as input and yields final category probabilities $\mathbf{p} \in \mathbb{R}^{|\mathcal{C}|+1}$ (covering $|\mathcal{C}|$ foreground classes plus a background class) with the corresponding bounding box regressions $\mathbf{b}$.

3.2. Overall Framework

The overall framework of our method is illustrated in Fig. 1, featuring a two-stage training paradigm that jointly regularizes feature learning through cross-head knowledge transfer and preserves classifier-level knowledge via prototype-based correction.

Specifically, Future-Aware Decoupled Cross-Head Distillation (FaCHD) decouples the classification head from the backbone and conducts bidirectional cross-head distillation. The ROI features from the incrementally trained student model are decoded by the classification heads of two frozen teacher models, producing complementary cross-head targets. The student's cross-predicted labels are then aligned with the teachers' outputs via a distillation loss, effectively mitigating the conflict between new-class supervision and old-class distillation. Building upon this decoupled head design, Region Prototype Semantic Correction (RPSC) further compensates for the semantic drift of multi-granularity region prototypes associated with old classes. By estimating feature-space misalignment between the old and new detectors, RPSC recalibrates old-class prototypes and integrates them with current prototypes to retrain the classification head, thereby reinforcing knowledge retention at the classification head level.

3.3. Future-Aware Decoupled Cross-Head Distillation

IOD requires the model to adapt to newly introduced categories while preserving discrimination of previously learned ones. However, the mismatch between new-class supervision and old-class distillation produces competing gradients on the shared classifier, leading to biased optimization and accelerated forgetting. To address this issue, FaCHD decouples the classification head from the backbone and introduces a dual-teacher cross-head distillation scheme that provides stable and consistent guidance across incremental tasks.

Specifically, the dual-teacher framework is composed of an old-class expert teacher $\mathcal{M}_{t-1}$ and a new-class intermediate teacher $\mathcal{M}_t^{im}$. $\mathcal{M}_{t-1}$ provides reliable guidance for preserving historical knowledge, while $\mathcal{M}_t^{im}$ is optimized only by $\mathcal{D}_t$, offering adaptive supervision aligned with new-class learning. The proposed regions from $\mathcal{M}_{t-1}$ serve as the distillation regions $\mathcal{R}$, ensuring that the knowledge transfer is concentrated on reliable old-class areas. Following the region partitioning and background label reconstruction strategy proposed in [29], we divide the distillation regions $\mathcal{R}$ according to their Intersection-over-Union (IoU) with the bounding boxes $\mathcal{Y}_t$ of new-class objects:

$$\begin{aligned} \mathcal{R}_1 &= \{\, \mathbf{d}_j \in \mathcal{R} \mid \forall \mathbf{y}_i \in \mathcal{Y}_t,\ \mathrm{IoU}(\mathbf{d}_j, \mathbf{y}_i) \le \lambda_2 \,\}, \\ \mathcal{R}_2 &= \{\, \mathbf{d}_j \in \mathcal{R} \mid \forall \mathbf{y}_i \in \mathcal{Y}_t,\ \mathrm{IoU}(\mathbf{d}_j, \mathbf{y}_i) > \lambda_2 \,\}, \end{aligned} \tag{1}$$

where $\mathcal{R}_1$ represents regions that are more likely to belong to old classes, while $\mathcal{R}_2$ corresponds to regions that are more likely to contain new classes.

Then, FaCHD includes a background probability reconstruction strategy [29] into the teacher model's target to construct a future-aware optimization target. Concretely, for regions in $\mathcal{R}_1$, the background probability from the old-class teacher is refined through the intermediate teacher, whereas for regions in $\mathcal{R}_2$, the background estimation of the intermediate teacher is corrected using the old-class teacher:

$$\begin{aligned} \hat{\mathbf{p}}_r^{c,im} &= \mathbf{p}_r^{c,im} \cdot \mathbf{p}_r^{b,t-1}, \quad r \in \mathcal{R}_1, \\ \hat{\mathbf{p}}_r^{c,t-1} &= \mathbf{p}_r^{c,t-1} \cdot \mathbf{p}_r^{b,im}, \quad r \in \mathcal{R}_2, \end{aligned} \tag{2}$$

where $\mathbf{p}_r^{c,\text{im}}$ represents probability distribution of the foreground class c generated by the intermediate teacher model $\mathcal{M}_t^{im}$ in the distillation region r, and $\mathbf{p}_r^{b,t-1}$ represents the background probability distribution estimated by the old teacher model $\mathcal{M}_{t-1}$ in the distillation region r.

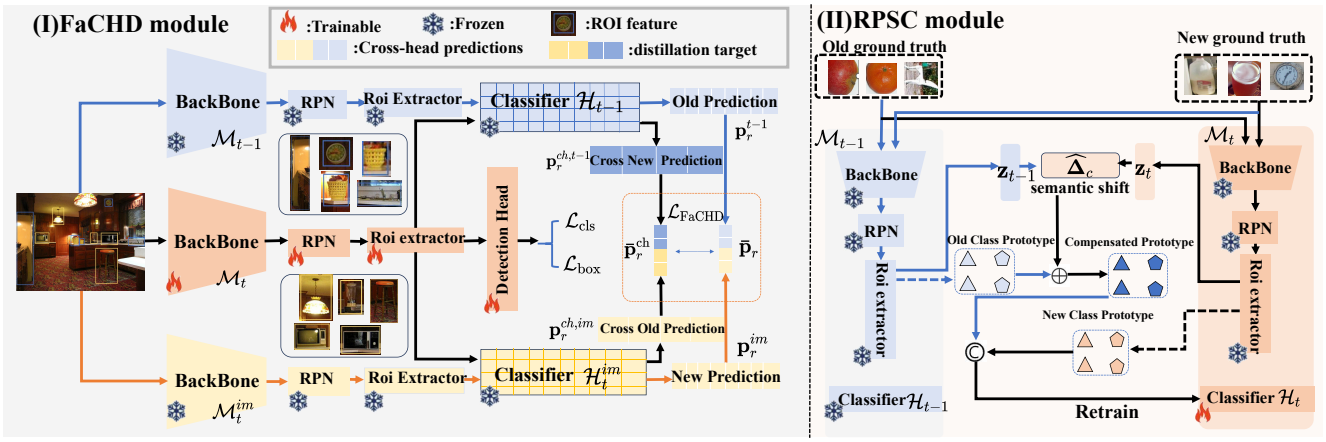


Figure 1. Overall framework of FaCHD-RPSC for incremental object detection. (I) The Future-Aware Decoupled Cross-Head Distillation (FaCHD) module is introduced, which decodes the ROI features of the student model using $\mathcal{M}_t$ and $\mathcal{M}_t^{im}$. This decouples the classification head of the student model and applies a distillation loss to regularize the backbone network. (II) The Region Prototype Semantic Correction (RPSC) module is introduced, which estimates the semantic drift of region prototypes by comparing feature maps of $\mathcal{M}_t$ and $\mathcal{M}_{t-1}$ on the current dataset. It compensates for the semantic drift of old class prototypes and retrains the classification head of $\mathcal{M}_t$ using the compensated old prototypes alongside the current prototypes.

Based on the reconstructed background probabilities, the teacher-side distillation target $\bar{\mathbf{p}}_r$ is constructed by concatenating the reconstructed background distribution with the original foreground class probabilities:

$$\bar{\mathbf{p}}_r = \begin{cases} \mathrm{concat}\big(\hat{\mathbf{p}}_r^{c,im},\ \mathbf{p}_r^{c,t-1}\big), & r \in \mathcal{R}_1, \\ \mathrm{concat}\big(\hat{\mathbf{p}}_r^{c,t-1},\ \mathbf{p}_r^{c,im}\big), & r \in \mathcal{R}_2, \end{cases} \tag{3}$$

where $\mathrm{concat}(\cdot,\cdot)$ denotes the concatenation of two probability label vectors along the rows.

To enable decoupled cross-head distillation, the ROI features $\mathbf{z}$ extracted from the student model are forwarded through the classification heads of both teacher models, producing cross-head predictions:

$$\mathbf{p}^{ch,t-1} = \mathrm{softmax}\big(\mathcal{H}_{t-1}(\mathbf{z})\big), \tag{4}$$

$$\mathbf{p}^{ch,im} = \mathrm{softmax}\big(\mathcal{H}_t^{im}(\mathbf{z})\big), \tag{5}$$

where $\mathbf{p}^{ch,t-1}$ represents the cross-head prediction obtained by decoding the ROI features of the student model through the classification head $\mathcal{H}_{t-1}$ of the old teacher model $\mathcal{M}_{t-1}$, and $\mathbf{p}^{ch,im}$ represents the cross-head prediction obtained by decoding the ROI features of the student model through the classification head $\mathcal{H}_t^{im}$ of the intermediate teacher model $\mathcal{M}_t^{im}$. Analogous to the teacher-side reconstruction, we reconstruct the background component of the student's cross-head predictions:

$$\begin{aligned} \hat{\mathbf{p}}_r^{ch,c,im} &= \mathbf{p}_r^{ch,c,im} \cdot \mathbf{p}_r^{ch,b,t-1}, \quad r \in \mathcal{R}_1, \\ \hat{\mathbf{p}}_r^{ch,c,t-1} &= \mathbf{p}_r^{ch,c,t-1} \cdot \mathbf{p}_r^{ch,b,im}, \quad r \in \mathcal{R}_2. \end{aligned} \tag{6}$$

Then, the reconstructed background probability of the cross-head predicted labels is connected with the original foreground probability distribution, resulting in the student model's joint cross-head predicted labels $\bar{p}_i^{ch}$:

$$\bar{\mathbf{p}}_r^{ch} = \begin{cases} \mathrm{concat}\big(\hat{\mathbf{p}}_r^{ch,c,im},\ \mathbf{p}_r^{ch,c,t-1}\big), & r \in \mathcal{R}_1, \\ \mathrm{concat}\big(\hat{\mathbf{p}}_r^{ch,c,t-1},\ \mathbf{p}_r^{ch,c,im}\big), & r \in \mathcal{R}_2. \end{cases} \tag{7}$$

Finally, the FaCHD distillation loss is defined as the mean Kullback–Leibler divergence between the future-aware teacher targets $\bar{\mathbf{p}}_r$ and the student's joint cross-head predictions $\bar{\mathbf{p}}_r^{\mathrm{ch}}$:

$$\mathcal{L}_{\mathrm{FaCHD}} = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \mathrm{KL}\big(\bar{\mathbf{p}}_r \,\|\, \bar{\mathbf{p}}_r^{\mathrm{ch}}\big). \tag{8}$$

The overall training objective combines the generic detection loss with the distillation term, weighted by a hyperparameter $\alpha > 0$:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{cls}} + \mathcal{L}_{\mathrm{box}} + \alpha\, \mathcal{L}_{\mathrm{FaCHD}}, \tag{9}$$

where $\mathcal{L}_{\mathrm{cls}}$ and $\mathcal{L}_{\mathrm{bbox}}$ denote the stander classification and bounding-box regression losses.

3.4. Region Prototype Semantic Calibration

Building upon the feature-level regularization achieved by FaCHD, RPSC further enhances knowledge preservation at the classifier level. RPSC calibrates multi-granularity region prototypes and re-trains the classification head after the distillation stage, ensuring consistent knowledge retention within the unified framework.

Given an input image $\mathbf{x}_i$, the backbone produces feature maps $\mathcal{M}_f(\mathbf{x}_i)$ and the RPN generates region proposals $\mathcal{P} = \mathcal{M}_{rpn}(\mathcal{M}_f(\mathbf{x}_i))$. For any proposal $r \in \mathcal{R}$, the

corresponding RoI feature $\mathbf{z}_i$ provides the region-level representation used throughout RPSC:

$$\mathbf{z}_i = \mathrm{RoIAlign}(r, \mathcal{M}_f(\mathbf{x}_i)), \tag{10}$$

where $\mathrm{RoIAlign}(\cdot,\cdot)$ performs a spatial alignment of the region proposal r by bilinearly interpolating the feature map into a fixed-size output. For each class c, a global region prototype is obtained as the mean of all RoI features:

$$\mu_c^g = \frac{1}{n_c}\sum_{i=1}^{n_c} \mathbf{z}_i^c, \tag{11}$$

where $\mathbf{z}_i^c$ represents the RoI feature for class c and μ_c^g denotes the global region prototype, obtained by computing the mean of all the n_c RoI features for class c.

Since global prototypes may overlook intra-class structural variations, we propose to reveal the intra-class local information that is difficult to capture by the global prototype. Following [39], the local prototypes are constructed by examining hypersphere neighborhoods in the feature space. Specially, for each RoI feature $\mathbf{z}_j^c$, we collect its cosine-similarity neighbors:

$$\mathcal{S}_j^c = \left\{ \mathbf{z}_i^c \;\middle|\; s_{i,j}^c > \gamma \right\}, \tag{12}$$

where $s_{i,j}^c = \cos(\mathbf{z}_j^c, \mathbf{z}_i^c)$, γ is a similarity threshold. Based on the number of ROI features within each hypersphere, the top-K hypersphere domains with the most features are greedily selected to obtain the most representative hypersphere, while ensuring that their centers do not overlap. We then compute the mean of the features within the selected hyperspheres to obtain a series of local region prototypes for class c:

$$\mu_c^\ell = \frac{1}{|\mathcal{S}_k^c|}\sum_{\mathbf{z}_i^c \in \mathcal{S}_k^c} \mathbf{z}_i^c, \qquad k = 1, \ldots, K, \tag{13}$$

where $\mathcal{S}_k^c$ denotes the set of hypersphere neighborhoods of class c containing the top-K features,and $\mathbf{z}_i^c$ represents the ROI features within the hypersphere neighborhood.

Incremental learning inevitably induces semantic drift, causing mismatches between old prototypes and the current feature space. Following SDC [43], RPSC estimates the drift by comparing RoI features extracted by the previous model $\mathcal{M}_{t-1}$ and the current model $\mathcal{M}_t$:

$$\boldsymbol{\delta} = \mathbf{z}^t - \mathbf{z}^{t-1}, \tag{14}$$

where $\mathbf{z}^t$ denotes the ROI feature map generated by the current model $\mathcal{M}_t$ and $\mathbf{z}^{t-1}$ represents the ROI feature map produced by the previous model $\mathcal{M}_{t-1}$.

Using the obtained difference in feature maps, we can compute the Gaussian affinity relative to the old class prototype and aggregate these offsets to measure the semantic drift for class c:

$$\widehat{\boldsymbol{\Delta}}_c = \frac{\sum \alpha_{i,c}\,\boldsymbol{\delta}}{\sum \alpha_{i,c}}, \quad \alpha_{i,c} = \exp\!\left(-\frac{\|\mathbf{z}^{t-1} - \mu_c^{t-1}\|^2}{2\sigma_c^2}\right), \tag{15}$$

where σ_c is the variance of the region prototypes for the old class c. Then, semantic drift compensation is performed for both the global and local prototypes of the old class. Specifically, the global and local prototypes of each old category are compensated in the following manner:

$$\begin{cases} \hat{\mu}_c^g = \mu_c^{g,(t-1)} + \widehat{\boldsymbol{\Delta}}_c, & c \in \mathcal{C}_{1:t-1}, \\ \hat{\mu}_c^\ell = \mu_c^{\ell,(t-1)} + \widehat{\boldsymbol{\Delta}}_c, & c \in \mathcal{C}_{1:t-1}. \end{cases} \tag{16}$$

For the prototypes of the new categories, since these prototypes are directly extracted from the current model, there is no semantic drift. Therefore, the global and local prototypes of the new categories remain unchanged:

$$\begin{cases} \widetilde{\mu}_c^g = \mu_c^{g,t}, & c \in \mathcal{C}_t, \\ \widetilde{\mu}_c^\ell = \mu_c^{\ell,t}, & c \in \mathcal{C}_t. \end{cases} \tag{17}$$

Prototype-conditioned classification head adaptation. After the distillation stage, we freeze the backbone network and the RPN to preserve the semantic feature distribution of the backbone network, ensuring that our prototypes consistently align with the current semantic feature distribution, and train the classification head independently using the region prototypes. For each old class, we concatenate the global prototype and the local prototype, denoted as μ^{t-1}. Similarly, for each new class, we concatenate the global prototype and the local prototype, denoted as μ^t.

Each prototype is fed into the RoI head's classification branch to obtain class probabilities:

$$\widehat{\mathbf{p}}_c^{t-1} = \mathrm{Softmax}\big(\mathcal{H}(\mu_{t-1,c})\big), \tag{18}$$

$$\widehat{\mathbf{p}}_c^{t} = \mathrm{Softmax}\big(\mathcal{H}(\mu_{t,c})\big), \tag{19}$$

where $\mathcal{H}(\cdot)$ denotes the process of feeding the prototype into the classification head to produce the corresponding prediction.

Finally, the classification head is optimized by a prototype-based replay loss:

$$\mathcal{L}_{\mathrm{re}} = -\sum_{c \in \mathcal{C}_{1:t-1}} \mathbf{y}_c \log\big(\widehat{\mathbf{p}}_c^{t-1}\big) - \lambda\, \mathbf{y}_c \sum_{c \in C_t} \log\big(\widehat{\mathbf{p}}_c^{t}\big), \tag{20}$$

where $\mathbf{y}_c$ denotes the ground truth label corresponding to the prototype,λ is the hyperparameter used to balance the replay loss between the new and old classes.

In practice, Upon completion of distillation, we rely solely on prototypes to update the parameters of the classification head: at each iteration, a prototype mini-batch is forwarded through $\mathcal{H}$, the loss in (20) is computed, and gradients are backpropagated to update only the classification head. The backbone and RPN remain frozen throughout this phase.

Table 1. PASCAL VOC 2007 single-step incremental results reported in mAP@0.5. The best number in each row is highlighted in **bold**; methods marked "$*$" leverage exemplars.

Methods	Venues	10-10			15-5			19-1			5-15		
		Old	New	All	Old	New	All	Old	New	All	Old	New	All
Joint Training	-	77.2	76.4	76.8	78.3	72.5	76.8	76.8	77.4	76.8	74.7	77.5	76.8
Fine-tuning	-	9.6	67.4	38.5	11.2	63.4	24.3	12.6	72.5	56.6	0.3	75.4	56.6
ORE* [15]	CVPR21	60.4	68.8	64.6	71.8	58.7	64.6	69.4	60.1	68.9	-	-	-
ILOD-Meta* [16]	TPAMI21	68.4	64.3	66.3	71.7	55.9	58.7	67.8	70.9	70.2	-	-	-
OW-DETR* [12]	CVPR22	63.5	67.9	65.7	72.2	59.8	69.1	70.2	62.0	69.8	-	-	-
ABR* [24]	ICCV23	71.2	72.8	72.0	73.0	65.1	71.0	71.0	69.7	70.9	64.7	71.0	69.4
GMDP-ABR* [38]	ICLR24	72.1	73.2	72.7	75.8	65.5	73.2	74.8	70.1	74.6	67.1	71.9	70.7
Faster ILOD [31]	NeurIPS15	69.8	54.5	62.1	71.6	56.9	67.9	68.9	61.1	68.5	62.0	37.1	43.3
PPAS [55]	Arxiv20	63.5	60.0	61.8	-	-	-	70.5	53.0	69.2	-	-	-
MMA [3]	CVPR22	69.3	63.9	66.6	73.0	60.5	69.9	71.1	63.4	70.7	66.8	57.2	59.6
ROSETTA [40]	CVPR22	67.6	66.0	66.8	71.7	61.6	69.8	69.7	**68.3**	69.6	-	-	-
PseudoRM [42]	MM23	69.1	68.6	68.9	73.4	60.9	70.3	72.9	67.3	72.6	-	-	-
PROB [47]	CVPR23	66.0	67.2	66.5	73.5	60.8	70.1	73.9	48.5	72.6	-	-	-
BPF [30]	ECCV24	71.7	74.0	72.9	75.9	63.0	72.7	74.5	65.3	74.1	66.4	75.3	73.0
GMDP-ILOD [37]	ICLR25	71.9	69.7	70.8	74.6	63.5	71.8	74.2	67.9	73.9	65.2	60.5	75.6
GDA-IOD [28]	ICCV25	75.1	74.6	74.9	77.1	63.2	73.6	76.1	58.4	75.2	69.2	75.8	74.1
Ours	-	**76.0**	**75.9**	**75.9**	**78.4**	**65.0**	**75.1**	**76.8**	58.5	**75.9**	**71.2**	**77.6**	**76.0**

Table 2. Quantitative comparison on the MS COCO benchmark. Methods employing rehearsal strategies (marked with $*$) are distinguished for fair comparison.

Methods	40+40			70+10		
	AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}
Joint Training	40.2	61.5	44.2	40.2	61.5	44.2
ILOD-Meta* [16]	23.8	40.5	24.4	-	-	-
ABR* [24]	34.5	57.8	35.2	31.1	52.9	32.7
NSGP-RePRE* [39]	35.4	55.3	38.6	36.5	56.0	39.8
Faster ILOD [31]	20.6	40.1	-	21.3	39.9	-
PseudoRM [42]	25.3	44.4	-	-	-	-
MMA [3]	33.0	**56.6**	34.6	30.2	52.1	31.5
BPF [29]	34.4	54.3	37.3	36.2	56.8	38.9
Ours	**35.5**	55.7	**38.9**	**36.9**	**57.1**	**40.1**

4. Experiments

4.1. Experimental Settings

Dataset and evaluation metrics. We conduct experiments on two widely adopted benchmarks for incremental object detection: Pascal VOC 2007 [9] and MS COCO 2017 [20]. The Pascal VOC 2007 dataset comprises 9,963 images annotated with bounding boxes spanning 20 object categories, while MS COCO 2017 contains over 118K images with annotations for 80 categories.

In the incremental learning setup, we denote a task sequence as "*A–B*", where the initial stage involves A classes and each subsequent step introduces B new classes. Following previous studies [4, 24, 29, 31, 41], categories are partitioned alphabetically to form non-overlapping subsets. Each image in dataset $\mathcal{D}_t$ contains at least one instance belonging to the corresponding category set $\mathcal{C}_t$.

For evaluation, we follow the standard protocol used in continual object detection. Consistent with prior work [42, 47], we report the mean Average Precision (mAP) computed across Intersection-over-Union (IoU) thresholds from 0.5 to 0.95, together with AP_{75} for COCO-based experiments. For Pascal VOC–based settings, performance is assessed using the AP_{50} metric. Notably, our method performs incremental object detection without employing exemplar replay, ensuring a fair comparison with recent approaches.

Implementation Details.. In line with prior studies, we develop our incremental object detection model based on Faster R-CNN [33] using a ResNet-50 backbone [14], initialized with weights pre-trained on ImageNet [7]. For the VOC (10-10) benchmark, we configure the hyperparameters as $\alpha = 20$ and $\lambda = 0.4$. All experiments on the VOC dataset are carried out on a single NVIDIA GeForce RTX 3090 GPU with a batch size of 16, utilizing PyTorch and the SGD optimizer.

4.2. Comparison with Sate-of-the-art Methods

As shown in Tables 1 and tab 3, our method outperforms existing approaches in both single-incremental and multi-incremental settings. In the challenging 19+1 setting, our

Table 3. The mAP@0.5 results across multiple incremental steps on the PASCAL VOC 2007 dataset are presented, with the best performance in each step highlighted in **bold**. Methods marked with "$*$" utilize exemplars.

Methods	Venues	10-5			5-5			10-2			15-1		
		1-10	11-20	1-20	1-5	6-20	1-20	1-10	11-20	1-20	1-15	16-20	1-20
Joint Training	-	76.9	76.0	76.4	73.6	77.4	76.4	76.9	76.0	76.4	78.3	70.7	76.4
Fine-tuning	-	5.3	30.6	18.0	0.5	18.3	13.8	3.8	13.6	8.7	0.0	10.5	5.3
ABR* [24]	ICCV23	68.7	67.1	67.9	64.7	56.4	58.4	67.0	58.1	62.6	68.7	56.7	65.7
GMDP-ABR* [38]	ICLR25	69.9	67.8	68.9	**66.3**	59.3	61.1	67.6	**58.9**	63.3	69.5	**58.9**	66.9
Faster ILOD [31]	NeurIPS15	68.3	57.6	63.1	55.7	16.0	25.9	64.2	48.6	56.4	66.9	44.5	61.3
MMA [3]	CVPR22	66.7	61.8	64.2	62.3	31.2	38.9	65.0	53.1	59.1	68.3	54.3	64.1
BPF [30]	ECCV24	69.1	68.2	68.7	60.6	63.1	62.5	68.7	56.3	62.5	71.5	53.1	66.9
GMDP-ILOD [37]	ICLR25	68.1	62.3	65.2	61.1	35.8	42.1	64.2	53.3	58.8	66.8	50.4	62.7
GDA-IOD [28]	ICCV25	71.3	67.2	69.3	62.2	63.9	63.5	-	-	-	72.4	50.4	67.2
Ours	-	**72.2**	**70.8**	**71.5**	63.2	**68.1**	**66.9**	**68.8**	57.1	**62.9**	**76.5**	41.3	**67.7**

Table 4. Ablation study on each component.

Idx	FaCHD	RPSC	VOC(10-10)			VOC(10-5)				VOC(5-5)				
			1-10	11-20	1-20	1-10	11-15	16-20	1-20	1-5	6-10	11-15	16-20	1-20
(a)	–	–	75.4	73.8	74.6	70.6	75.2	59.2	68.9	47.2	64.0	76.3	56.4	61.0
(b)	✓	–	75.7	75.0	75.4	71.4	78.3	61.7	70.7	62.8	66.0	74.8	58.6	65.5
(c)	–	✓	75.8	74.0	74.9	70.7	76.6	61.4	70.0	49.4	63.2	75.8	57.1	61.4
(d)	✓	✓	**76.0**	**75.9**	**75.9**	**72.2**	**78.4**	**63.0**	**71.5**	**63.2**	**67.0**	**76.7**	**60.6**	**66.9**

results reached 75.9%, surpassing GMDP-ILOD [37] by 2%. In the long-duration multi-incremental setting, such as 5-5, our method achieved 66.9%, outperforming BPF [29] and GMDP-ABR [24] by 4.4% and 5.8%, respectively, demonstrating the superiority of our approach in long-term incremental learning. Notably, in the 10-10 setting, our method improves the mAP on old and new tasks by 0.9% and 1.3%, respectively, compared to GDA-IOD [28]. This indicates that our approach achieves better stability without sacrificing plasticity.

For the MS COCO 2017 dataset, we present the incremental results for the first 40 + last 40 and first 70 + last 10 settings in Table 2. By applying FaCHD to impose explainable constraints on the detector region features for both old and new tasks, and utilizing ROI prototype replay with semantic shift compensation to retain knowledge from previous tasks in the classification head, GDA-IOD consistently surpasses rehearsal-free methods. Specifically, in the 40+40 and 70+10 scenarios, our method achieved the highest AP of 35.5% and 36.9%, respectively, demonstrating its superior performance.

4.3. Ablation Study

We conduct extensive ablation studies to validate the effectiveness of the key components in our method.

Contributions of individual components. As shown in Tab 4, we progressively add each component under the VOC 10–10, VOC 10–5, and the more challenging VOC 5–5 settings. We first reproduce a baseline model (a) that adopts a pseudo-labeling strategy augmented with a Gaussian Mixture Model [28]. The results in Tab. 4 demonstrate that the proposed FaCHD is effective across incremental settings. Under the 10-10 setting, our method achieves a 1.2% performance gain on classes 11–20, while maintaining strong retention of knowledge from previously learned classes. This indicates that decoupling the student's ROI features using two frozen teachers, the old model and an intermediate model, and distilling them with a background-fusion strategy allows FaCHD to both retain old-class knowledge and maintain robust learning on new classes.In the 10–5 scenario, FaCHD boosts mAP by 1.8%, and in the 5–5 scenario it achieves a 4.5% gain. Moreover, compared with model (b), model (d)—which integrates ROI prototypes with semantic-drift compensation—consistently performs better across all incremental settings. These results suggest that our drift-compensated ROI prototypes, coupled with retraining of the classification head, effectively preserve memory of old classes and mitigate catastrophic forgetting.

Comparison with DwF. To validate the effectiveness

Table 5. Ablation study between FaCHD and DwF.

Methods	10-10			5-15			19-1			10-5			
	1-10	11-20	1-20	1-5	6-20	1-20	1-19	20	1-20	1-10	11-15	16-20	1-20
BaseLine	75.4	73.8	74.6	68.2	76.3	74.2	74.9	62.9	74.3	70.6	75.2	59.2	68.9
DwF++	75.2	74.9	75.0	69.1	76.9	74.9	76.2	52.3	75.0	71.5	76.8	60.9	70.2
FaCHD	75.7	75.0	75.4	69.0	77.5	75.4	77.1	46.0	75.6	71.4	78.3	61.7	70.7

of the FaCHD method in decoupling the classification head and performing distillation, we compare FaCHD with DwF [29]. Since the dynamic pseudo-label strategy has not been widely used in previous studies, we introduce the dynamic pseudo-label strategy into DwF for a fair comparison, denoted as DwF++. The DwF method directly computes the distillation loss by connecting the predicted labels of the student model with the reconstructed background probabilities of the teacher model, without considering the potential conflict between the new class supervisory signal and the distillation supervisory signal on the classification head. In contrast, our FaCHD method decouples the classification head, with the distillation loss primarily regularizing the backbone network. Under four different settings, our method consistently outperforms DwF++ by 0.4%,0.5%, 0.6%, and 0.5% in mAP, respectively, highlighting the superiority of decoupled cross-head distillation over the DwF method in incremental object detection.

Hyper-parameter λ and α..We analyzed the impact of two key hyperparameters, α and λ. The hyperparameter α in Eq.(9) controls the strength of distillation during training. As shown in Figure 2, the best performance in the VOC10-10 incremental setting occurs when α is set to 20. Notably, as α increases from 5 to 20, both the mAP for old and new classes improve, indicating that our method not only helps the model retain knowledge of old classes but also provides good plasticity for learning new classes. When α exceeds 20, performance starts to decline, which we hypothesize is due to $\alpha > 20$ hindering the model's convergence.

Additionally, we analyzed the hyperparameter λ Eq.(20), which controls the loss ratio between the new and old region prototypes involved in retraining the model's classification head. As shown in Figure 2, our model achieves the best performance on VOC10-10 when λ is set to 0.4. Although the impact of different λ values on performance varies slightly, the model overall demonstrates strong robustness.

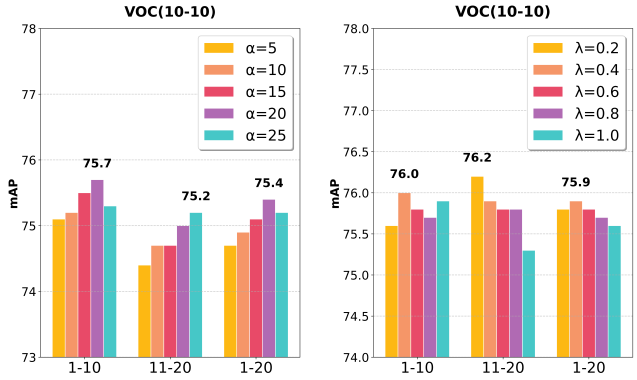


Figure 2. Ablation experiments with varying α in Eq. (9) and λ in Eq (20) on VOC 10-10.

5. Conclusion

In this work, we address the core challenges of incremental object detection by mitigating the conflict between new-task supervision and distillation signals in the classification head, while simultaneously improving knowledge retention. We introduce Future-Aware Decoupled Cross-Head Distillation, which decouples the student model's classification head and regularizes the backbone through targeted distillation. To further preserve knowledge of previously learned classes, we propose Region Prototype Semantic Calibration, a mechanism that estimates semantic drift by comparing feature-map discrepancies across models and compensates for outdated region prototypes. These calibrated prototypes enable effective deduplicated training of the classification head. In future research, we plan to explore more flexible prototype modeling that adapts continuously across tasks, extend our framework to long-tail and open-vocabulary detection scenarios, and investigate cross-modal or self-supervised signals to further enhance robustness under large domain or appearance shifts.

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